

Book 7: DevOps & Cloud Infrastructure for Banking

Docker Multi-Stage, Kubernetes & GCP Enterprise Architecture

IDFC FIRST Bank | Strategic Projects Developer (3+ Years Experience)

Personalized Hinglish Interview Preparation System • Production Ready

Docker Containerization, Multi-Stage Builds & Graceful Shutdown

1. Why This Matters

At IDFC FIRST Bank's Strategic Projects, all microservices run containerized inside Kubernetes (GKE / EKS / OpenShift). In a technical interview, writing a basic `FROM node:latest` Dockerfile signals inexperience with production systems. Senior engineers evaluate whether you know how to build **secure, minimal multi-stage Docker images**, run containers as **non-root users**, and handle **OS process signals (`SIGTERM`)** to ensure financial transactions in flight are not abruptly severed during deployments.

2. Prerequisites

- Linux processes, signals (`SIGINT` , `SIGTERM` , `SIGKILL`).
 - Basic Docker commands (`docker build` , `docker run`).
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3. Concept

Multi-Stage Docker Builds

A naive Dockerfile packages the entire source tree, compiler toolchains (`gcc` , `python` , `make`), development dependencies (`devDependencies`), and package managers into the final image. This results in: - Bloated images ($> 1\text{ GB}$). - Severe security vulnerabilities (attackers have access to compilers and package managers inside the container).

```

flowchart LR
    subgraph Stage1["Stage 1: Build & Compile (Fat Builder)"]
        BaseNode["node:20-alpine\nIncludes build tools, TypeScript, devDeps"]
        Compile["npm run build\nProduces compiled JS in /dist"]
    end

    subgraph Stage2["Stage 2: Production Runtime (Lean Image)"]
        MinimalNode["node:20-alpine (Clean)"]
        CopyDist["Copy ONLY /dist & production node_modules"]
        NonRoot["Switch to non-root user: node"]
        CMD["CMD ['node', 'dist/server.js']"]
    end

    BaseNode --> Compile
    Compile -->|Artifact Transfer| CopyDist
    MinimalNode --> CopyDist --> NonRoot --> CMD

```

4. Simple Example: Naive vs Multi-Stage Dockerfile

Naive Dockerfile (Dangerous & Bloated):

```

# NEVER DO THIS IN PRODUCTION:
FROM node:20
WORKDIR /app
COPY . .
RUN npm install
RUN npm run build
EXPOSE 3000
CMD ["npm", "start"] # Runs via npm wrapper which swallows SIGTERM!

```

5. Production Banking Dockerfile: Secure Multi-Stage Build

```
# =====  
# STAGE 1: Builder  
# =====  
FROM node:20-alpine AS builder  
  
WORKDIR /app  
  
# Install dependencies needed for compiling native C++ addons  
RUN apk add --no-cache python3 make g++  
  
# Copy package manifests first to leverage Docker layer caching  
COPY package*.json tsconfig.json ./  
  
# Install ALL dependencies (including devDependencies for building)  
RUN npm ci  
  
# Copy application source code  
COPY src/ ./src  
  
# Compile TypeScript to JavaScript  
RUN npm run build  
  
# Prune devDependencies, leaving only production modules  
RUN npm prune --production  
  
# =====  
# STAGE 2: Production Runner  
# =====  
FROM node:20-alpine AS runner  
  
WORKDIR /app  
  
# Set production environment  
ENV NODE_ENV=production  
  
# Install dumb-init or tini to handle PID 1 signal forwarding  
RUN apk add --no-cache dumb-init  
  
# Copy only production artifacts from builder stage  
COPY --from=builder /app/package*.json ./  
COPY --from=builder /app/node_modules ./node_modules  
COPY --from=builder /app/dist ./dist  
  
# Security: Never run containers as root!  
USER node  
  
EXPOSE 3000
```

```
# Use dumb-init as entrypoint to properly forward OS signals
ENTRYPOINT ["dumb-init", "--"]

# Execute Node directly (avoid npm start which swallows signals)
CMD ["node", "dist/server.js"]
```

6. Code: Graceful Shutdown Implementation in Express / Node.js

When Kubernetes scales down a pod or deploys a new version, it sends a `SIGTERM` signal. Your application must:

1. Stop accepting new incoming HTTP connections.
2. Allow currently in-flight financial transfers to finish processing.
3. Close database connection pools cleanly.
4. Exit process with code 0.

```
const express = require('express');
const http = require('http');
const { pool } = require('./db');

const app = express();
const server = http.createServer(app);

// Liveness Probe (Is container alive?)
app.get('/healthz', (req, res) => res.status(200).send('OK'));

// Readiness Probe (Is container ready to serve traffic?)
let isShuttingDown = false;
app.get('/ready', (req, res) => {
  if (isShuttingDown) {
    return res.status(503).send('Service Shutting Down');
  }
  return res.status(200).send('Ready');
});

// Start Server
const PORT = process.env.PORT || 3000;
server.listen(PORT, () => console.log(`Payment Service listening on port ${PORT}`));

// Graceful Shutdown Handler
function handleGracefulShutdown(signal) {
  console.log(`Received ${signal}. Initiating graceful shutdown...`);
  isShuttingDown = true;

  // 1. Stop accepting new connections
  server.close(async () => {
    console.log('HTTP server closed. In-flight requests finished.');
```

```
    try {
      // 2. Drain and close database connection pool
      console.log('Closing database connection pool...');
      await pool.end();
      console.log('Database connections closed cleanly.');
```

```
      process.exit(0);
    } catch (err) {
      console.error('Error during database pool shutdown:', err);
      process.exit(1);
    }
  });

  // Force exit if shutdown hangs beyond 25 seconds (K8s default grace period is 30s)
  setTimeout(() => {
```

```

        console.error('Graceful shutdown timeout exceeded! Forcefully
terminating.');
```

```

        process.exit(1);
    }, 25000).unref();
}

// Listen for Kubernetes termination signals
process.on('SIGTERM', () => handleGracefulShutdown('SIGTERM'));
process.on('SIGINT', () => handleGracefulShutdown('SIGINT'));
```

7. How It Works Internally: The PID 1 Problem in Linux Containers

In Linux, **Process ID 1 (PID 1)** is the init process (`systemd`). - PID 1 has special kernel properties: default signal handlers are not registered for it. - If you start a container with `CMD ["npm", "start"]` or `CMD ["node", "server.js"]` without an init process, Node runs as PID 1. - When Kubernetes sends `SIGTERM`, Node.js may ignore it, or `npm` may consume the signal without forwarding it to the child Node process! - After the Kubernetes termination grace period (30s), the kernel sends `SIGKILL`, abruptly aborting active payment transactions mid-flight. - **Solution:** Use `dumb-init` or `tini` as the container entrypoint. It runs as PID 1, spawns Node as a child process, and properly forwards all OS signals (`SIGTERM`, `SIGINT`) while reaping orphaned zombie processes.

8. Common Mistakes

1. **Running Containers as `root`** : By default, Docker containers run as the root user (`UID 0`). If an attacker exploits an RCE vulnerability in an npm package, they have root access to the underlying container namespace. Always switch to `USER node` .
2. **Using `npm start` in `CMD`** : `npm start` spawns a sub-shell that swallows POSIX termination signals, breaking graceful shutdown. Always use `CMD ["node", "server.js"]` .
3. **Copying `node_modules` from Host Machine**: Always put `node_modules` in `.dockerignore` . Copying local modules into a Linux container fails because native C++ binaries (e.g., `bcrypt`) built on macOS or Windows cannot run on Alpine Linux.

9. Performance / Complexity Matrix

Image Strategy	Image Size	Build Time	Security Posture
<code>node:latest</code> (Single Stage)	$\approx 1.1\text{ GB}$	Fast	Dangerous (Includes compilers, root user)
<code>node:alpine</code> (Single Stage)	$\approx 350\text{ MB}$	Moderate	Moderate
Multi-Stage with <code>node:alpine</code> + <code>USER node</code>	$\approx 120\text{ MB}$	Optimized	Production & Banking Grade

10. Interview Questions (Easy \$to\$ Medium \$to\$ Hard)

Easy

- **Q:** Why should you avoid using `node:latest` in production Dockerfiles?

Medium

- **Q:** Explain how Docker layer caching works. Why do we copy `package.json` and run `npm install` before copying the rest of the application code?

Hard

- **Q:** Describe the "PID 1 problem" in Docker containers. What happens when a containerized Node.js application receives a `SIGTERM` from Kubernetes, and how does `dumb-init` or `tini` solve it?

11. Follow-up Questions from Interviewer

- "What is the difference between Kubernetes Liveness and Readiness probes in a banking microservice?" (Answer: Liveness checks if the process is alive; if it fails, K8s restarts the container. Readiness checks if the process can accept traffic; if it fails, K8s removes the pod from the service load balancer without killing it).
- "What is a Distroless Docker image and why is it increasingly mandated in banking institutions?"

12. Model Answer: Docker Layer Caching Strategy

Interviewer: "Why do we split the `COPY` instructions for `package.json` and the source code in a `Dockerfile`?"

Model Answer: "Docker creates an immutable cached layer for each instruction in a `Dockerfile`. A cache hit occurs only if the instruction and all underlying files have not changed since the previous build.

Running `npm ci` is expensive because it downloads and verifies hundreds of packages.

- If we write `COPY . .` followed by `RUN npm ci`, every tiny modification to a JavaScript file, comment, or `README` invalidates the cache from that point onward, forcing a full `npm ci` re-download on every single build.
- By copying `package*.json` first and executing `RUN npm ci`, Docker caches the `node_modules` layer. Subsequent code changes only invalidate the later `COPY src/ ./src` layer.
- As a result, code-only continuous integration builds take **3–5 seconds** instead of minutes, accelerating CI/CD deployment pipelines."

13. Practical Exercise

Review the multi-stage `Dockerfile` in Section 5. Create a `.dockerignore` file in your mental model:

```
node_modules
dist
.git
.env
*.log
```

14. Quick Revision

- Multi-stage builds separate compilation tools from the lean runtime container.
 - Always run containers as non-root (`USER node`).
 - Handle `SIGTERM` to close HTTP listeners and drain database connections before exiting.
 - Use `dumb-init` or `tini` to solve PID 1 signal forwarding and zombie reaping.
 - Leverage Docker layer caching by copying package manifests before source code.
-

15. Interview Checklist

- ☐ Can write a 2-stage Dockerfile from memory.
- ☐ Explains why containers should never run as root.
- ☐ Understands the PID 1 problem and signal forwarding.
- ☐ Writes graceful shutdown logic handling `SIGTERM` .

Google Cloud Platform (GCP) Infrastructure for Enterprise Banking

1. Why This Matters

IDFC FIRST Bank leverages modern cloud and hybrid infrastructure to scale digital acquisition, mobile banking APIs, and analytics. As a 3+ YoE developer with familiarity in GCP, demonstrating how to architect secure, highly available services using **Google Kubernetes Engine (GKE)**, **Cloud Run**, **Cloud SQL**, **Cloud Pub/Sub**, and **IAM Workload Identity** proves that you understand modern production cloud infrastructure, rather than just writing code locally on `localhost:3000`.

2. Prerequisites

- Basic networking: Virtual Private Cloud (VPC), Subnets, CIDR blocks, NAT Gateways.
 - Cloud primitives: Virtual Machines, Containers, Object Storage, IAM.
-

3. Concept: Enterprise Banking VPC Architecture on GCP

```

flowchart TD
    Internet["Public Internet / Mobile Clients"] --> CloudArmor["Cloud Armor (WAF & DDoS Mitigation)"]
    CloudArmor --> ExtLB["External HTTPS Global Load Balancer"]

    subgraph VPC["Bank Private VPC (Restricted Mumbai & Delhi Regions - RBI Compliant)"]
        ExtLB --> PublicSubnet["Public Ingress Subnet"]
        PublicSubnet --> IntLB["Internal Layer 7 Load Balancer"]

        subgraph AppSubnet["Application Subnet (Private IP Only - No Public IPs)"]
            IntLB --> GKECluster["GKE / Cloud Run Private Cluster\nPayment & Ledger Microservices"]
        end
    end

    subgraph DataSubnet["Data Subnet (Private Service Connect)"]
        GKECluster --> CloudSQL["Cloud SQL (PostgreSQL Multi-AZ Primary)"]
        CloudSQL --> Replica["Cloud SQL Read Replica"]
        GKECluster --> Memorystore["Memorystore for Redis (Cluster)"]
        GKECluster --> PubSub["Cloud Pub/Sub (Event Bus)"]
    end

    subgraph SecurityPlane["Security & Management Plane"]
        GKECluster --> SecretMgr["Secret Manager (DB Credentials)"]
        GKECluster --> CloudKMS["Cloud KMS (Customer-Managed Encryption Keys - CMEK)"]
    end

    NAT["Cloud NAT Gateway\n(Allows outbound bank updates without inbound public IP)"]
    AppSubnet --> NAT
end

```

4. Key GCP Services in Digital Banking

1. **Google Kubernetes Engine (GKE):** Managed Kubernetes for containerized microservices. In banking, GKE is configured with **Private Clusters** (nodes have no public IP addresses; API server endpoint is restricted to authorized networks).
2. **Cloud Run:** Fully managed serverless container runtime. Ideal for lightweight event-driven microservices (e.g., PDF generation, statement dispatch, webhook ingestion) that scale to zero when idle.
3. **Cloud SQL for PostgreSQL:** Managed relational database with automated High Availability (HA) across multiple Availability Zones, automated point-in-time recovery (PITR), and automatic WAL

archiving.

4. **Cloud Armor:** Web Application Firewall (WAF) and DDoS protection that blocks OWASP Top 10 vulnerabilities, enforces rate limiting, and geofences inbound traffic strictly to Indian IP ranges.
 5. **Cloud KMS (Key Management Service):** Stores encryption master keys in FIPS 140-2 Level 3 Hardware Security Modules (HSMs) for **Customer-Managed Encryption Keys (CMEK)**.
 6. **Secret Manager:** Stores API keys, database credentials, and certificates with strict versioning and automatic rotation.
-

5. Real-World Banking Example: Workload Identity Federation (Eliminating Service Account Keys)

In traditional cloud setups, developers downloaded a JSON service account key (`service-account-key.json`) and mounted it inside the container. - **The Security Danger:** If an engineer accidentally commits the JSON file to GitHub or a developer workstation is compromised, attackers gain permanent administrative access to the bank's cloud resources. - **The Banking Standard (Workload Identity):** - GKE maps a native Kubernetes Service Account (KSA) directly to a Google Cloud IAM Service Account (GSA). - When the Node.js application calls GCP APIs (Cloud Storage or Secret Manager), the Google Cloud Client Library retrieves short-lived, automatically rotated OIDC tokens from the local metadata server (`http://metadata.google.internal`). - **Zero static keys exist anywhere on disk or in environment variables!**

6. Code: Accessing GCP Secret Manager Securely in Node.js

```

const { SecretManagerServiceClient } = require('@google-cloud/secret-manager');
const client = new SecretManagerServiceClient();

/**
 * Retrieves database credentials dynamically from GCP Secret Manager
 * at startup using Workload Identity (no static credentials needed!).
 */
async function getDatabaseCredentials(projectId, secretName, version = 'latest') {
  const name = `projects/${projectId}/secrets/${secretName}/versions/${version}`;

  try {
    const [response] = await client.accessSecretVersion({ name });
    const payload = response.payload.data.toString('utf8');
    return JSON.parse(payload);
  } catch (err) {
    console.error(`Failed to access secret ${secretName}:`, err.message);
    throw err;
  }
}

// Usage in application startup:
(async () => {
  const dbConfig = await getDatabaseCredentials('idfc-banking-prod', 'database-master-credentials');
  console.log('Successfully fetched credentials for user:', dbConfig.username);
})();

```

7. How It Works Internally: Private Service Connect & Cloud NAT

In an enterprise banking VPC: 1. **No Public IP Addresses:** Worker nodes in the GKE cluster must have **zero public IPv4 addresses**. 2. **Private Service Connect (PSC):** Connections from GKE to Cloud SQL or Cloud Storage traverse Google's private software-defined network backbone via private IP endpoints, completely bypassing the public internet. 3. **Cloud NAT:** When a microservice needs to initiate an outbound connection to an external partner bank or NPCI switch, it routes through **Cloud NAT**. The external switch sees a fixed, whitelisted static public IP, while no inbound connections can penetrate back through the NAT.

8. Common Mistakes

1. **Hardcoding Secrets in Environment Variables or Dockerfiles:** Environment variables leak through crash dumps, `docker inspect`, and process listings. Always fetch secrets at runtime via Secret Manager or mount them as Kubernetes Secret volumes backed by Secret Store CSI Driver.

2. **Deploying Cloud Resources Outside India:** For IDFC FIRST Bank, creating Cloud SQL or GCS buckets in `us-central1` or `eu-west1` is a direct violation of RBI Data Localization mandates. All resources must be provisioned in `asia-south1` (Mumbai) or `asia-south2` (Delhi).
3. **Overly Permissive IAM Roles:** Granting `Editor` or `Owner` roles to service accounts violates the **Principle of Least Privilege**. Service accounts should only hold narrow roles (e.g., `roles/secretmanager.secretAccessor` on a specific secret).

9. Performance / Complexity Matrix

GCP Service	Compute / Storage Model	High Availability SLA	Best Used For
GKE Private Cluster	Kubernetes Container Orchestration	\$99.95\%\$ (Zonal) / \$99.99\%\$ (Regional)	Core transactional microservices
Cloud Run	Serverless Container	\$99.95\%\$	Event-driven APIs & Webhook receivers
Cloud SQL	Managed PostgreSQL Multi-AZ	\$99.95\%\$	Transaction ledger & relational source of truth
Cloud Storage (GCS)	Object Storage (Regional)	\$99.99\%\$	PDF statements & raw settlement file archive
Cloud Pub/Sub	At-least-once Distributed Queue	\$99.95\%\$	Transactional events & notification triggers

10. Interview Questions (Easy \$to\$ Medium \$to\$ Hard)

Easy

- **Q:** What is the difference between a regional and zonal resource in Google Cloud?

Medium

- **Q:** Why should production Kubernetes nodes in a banking deployment be configured as a "Private Cluster" without public IP addresses? How do pods access external partner APIs?

Hard

- **Q:** Explain how Workload Identity works in GKE. Walk through the authentication handshake between a Kubernetes pod, the GKE metadata server, and Google Cloud IAM.

11. Follow-up Questions from Interviewer

- "How does Cloud Armor mitigate a Layer 7 HTTP flood attack targeting an IDFC login endpoint?"
(Answer: Enforces adaptive rate-limiting rules based on IP or session tokens, validates reCAPTCHA Enterprise tokens, and inspects HTTP request headers for known malicious bot signatures).
 - "What is Customer-Managed Encryption Keys (CMEK) and why does RBI mandate it over default Google-managed keys?"
-

12. Model Answer: Workload Identity Federation

Interviewer: "How do you securely grant a Node.js microservice running in GKE access to GCP Cloud SQL and Secret Manager without storing static credentials?"

Model Answer: "We implement **GKE Workload Identity**, which is Google Cloud's recommended best practice for zero-trust security.

1. We create a Google Cloud IAM Service Account (GSA) and grant it the exact least-privilege IAM roles needed (e.g., `roles/secretmanager.secretAccessor`).
2. We create a Kubernetes Service Account (KSA) in our namespace and annotate it with the GSA's email address.
3. We bind the two accounts using IAM policy binding (`roles/iam.workloadIdentityUser`).

At Runtime: When our Node.js app initializes the Google Cloud Client SDK, the SDK queries the local GKE node metadata server.

The GKE metadata server intercepts the call, validates the pod's projected Kubernetes service account token, exchanges it with the Google Security Token Service (STS) for a short-lived Google IAM access token (valid for 1 hour), and returns it to the client SDK.

As a result, our code accesses Cloud SQL and Secret Manager seamlessly without any static credentials, private keys, or passwords ever residing in configuration files, Git repositories, or container images."

13. Practical Exercise

Review how your Node.js application configures database connection strings: ensure it supports loading connection configurations from environment variables populated via Kubernetes Secrets or runtime secret fetches rather than static config files.

14. Quick Revision

- All banking cloud resources must reside in Indian regions (`asia-south1` / `asia-south2`) for RBI compliance.
 - GKE Private Clusters prevent worker nodes from having public IP addresses.
 - Cloud NAT enables secure outbound traffic without exposing inbound ports.
 - Workload Identity eliminates static JSON service account keys.
 - Cloud Armor protects banking APIs against DDoS and OWASP Layer 7 threats.
-

15. Interview Checklist

- ☐ Understands the difference between GKE and Cloud Run.
- ☐ Explains the architecture of a Private Cluster and Cloud NAT.
- ☐ Articulates how Workload Identity functions under the hood.
- ☐ Knows the RBI data localization requirement for GCP regions.